JUAS2 2026: Normal-Conducting Magnet Case Study

Alastair Gill[1], *Colin Rigou*[2], *Leo Bar Grossman*[3], *Margherita Morriello*[4], *Maria Vagianni*[5], *Santiago Martinez*[6]
[1] CERN, Geneva, Switzerland
[2] Grenoble INP Phelma, UGA, Grenoble, France
[3] Technische Universität Berlin, Berlin, Germany
[4] Università degli Studi di Padova, Padova, Italy
[5] Université Paris Saclay (GI PLATO), France
[6] Université Paris Saclay (LASCALA), France

Abstract
This report presents the analytical and numerical preliminary design of a normal-conducting dipole magnet for a transfer-line application. The study combines magnetic sizing, conductor selection, electrical checks under converter limits, and cooling-circuit calculations based on thermal and hydraulic constraints.

Keywords
normal-conducting magnets; dipole magnet; analytical design; cooling circuit; FEMM.

1 Introduction

The goal of this report is the analytical and numerical design of the a dipole magnets for the transfer line of MedAustron, the required magnets will be 3 magnets (plus an additional spare magnet). The study combines first-order analytical sizing, magnetic, electrical, and thermal cooling constraints with the definition of the geometric and excitation parameters required for subsequent numerical validation in FEMM software.

MedAustron is a synchrotron-based hadron therapy and research facility located in Wiener Neustadt, approximately 50 km south of Vienna, next to the future site of the new hospital. The facility is dedicated to medical treatment and non-clinical research.

The accelerator complex is based on a synchrotron with a circumference of approximately 76 m and operates with two ion species: protons and carbon ions. The relevant beam energy ranges are 60 to 220 MeV for protons (medical operation) and 120 to 400 MeV/n for carbon ions.

Within this context, the dipole magnets must satisfy beam optics and field quality requirements while remaining compatible with the available mechanical envelope, power-converter limits, and cooling constraints. The present report focuses on deriving a feasible design solution and preparing the corresponding inputs for the numerical model.

2 Input Specifications and Constraints

This section summarizes the fixed inputs that define the design space of the transfer-line dipoles.

In addition to the numerical requirements above, the design is constrained by several fixed implementation choices. The magnet is operated in quasi-static (DC) mode, which directly affects the electrical model and converter-voltage estimate through the dominance of the resistive term with a limited inductive contribution during slow ramps. The yoke material is fixed to laminated steel M1200-100A (available on stock), which constrains the magnetic design and saturation margin to a maximum of 1.5T. The three

Table 1: Functional specification for the bending dipoles.

Parameter	Value
Bending angle (per magnet)	36°
Maximum mechanical length per magnet l_{mech}	0.9 m
Horizontal good-field region	±30 mm
Vertical good-field region	±23 mm
Field quality inside GFR $\Delta B/B_0$	$< \pm 1 \times 10^{-3}$
Available cooling-water pressure P_{in}	10 bar
Required return pressure P_{out}	2 bar
Inlet water temperature T_{in}	20 °C
Maximum temperature rise on magnet ΔT	15 °C
Maximum converter current I_{max}	600 A
Maximum converter voltage (3 magnets in series) V_{max}	100 V

Table 2: Beam and lattice parameters used for the preliminary sizing.

Parameter	Value	Unit
Beam energy	7	MeV/u
Beamline length	40.9	m
Beta function β_x, β_y	≈ 10	m
Beam size σ_x, σ_y	±10	mm

magnets must be electrically connected in series because only one power converter is available, which makes the total converter voltage the relevant electrical constraint. One complete spare unit is required, which affects conductor stock planning. Finally, the preferred choice is to use one of the available hollow OF-Cu conductors already in market, in order to reduce procurement effort and align the design with available manufacturing resources.

The available conductor options are hollow OF-Cu bars with outer dimensions and cooling-duct diameters as specified by the stock list (Types 1-4). These options are treated as discrete design candidates in the analytical sizing, and their impact is assessed through current density, coil resistance, voltage drop, dissipated power, and cooling performance.

3 Parameters to be determined

The purpose of the analytical design is to determine a consistent dipole solution that satisfies the specifications and constraints stated above. The quantities to be determined are not listed in a table, since they are design outputs including but not limited to numerical inputs and are better presented as part of the design logic.

The analytical design must determine the magnet topology (straight or curved) and the magnetic-circuit layout (H-shape or C-shape), followed by the main geometric quantities: aperture height h, yoke length l_{yoke}, effective magnetic length l_{eff}, pole width w, and return-yoke thickness. Then, the operating flux density B and the required excitation NI (ampere-turns) must be established, from which the nominal current I and number of turns N are selected under the converter-current constraint.

Once the excitation is fixed, the conductor candidate is selected by evaluating conductor dimensions, current density J, and coil envelope dimensions (including conductor length and insulation allowance). The corresponding electrical quantities such as coil resistance R, voltage drop V, and dissipated power P_Ω, are then computed and checked against the converter voltage limit for the three magnets connected in series.

Finally, the cooling design is derived from the dissipated power and thermal limits. This includes the coolant flow rate Q, required pressure drop ΔP, average coolant velocity v_{avg}, and Reynolds number Re, which together determine whether the selected conductor and cooling duct provide a feasible operating point.

3.1 Parameters Required for the FEMM Model

The analytical design also provides the geometric and excitation inputs needed to build the numerical FEMM model. In practice, this includes the aperture height, pole width, yoke dimensions, coil-window dimensions, coil dimensions, coil position, and coil excitation (ampere-turns), together with a consistent geometry sketch defined by key-point coordinates (x, y) with respect to the beam axis. These parameters are generated from the analytical sizing and then used for field validation and refinement in the numerical model.

The next section presents the analytical design procedure and computes these quantities under the constraints defined in this section.

4 Analytical design of the Magnet

4.1 Magnet Type

4.1.0.1

Choice: Straight magnet.

A straight dipole configuration is selected for the present design. The expected beam sagitta over the magnet length is small enough that a curved geometry would not provide a significant benefit in terms of magnetic material usage or aperture optimization. In contrast, a straight magnet is simpler and less expensive to manufacture, since it avoids custom tooling and reduces machining and assembly complexity.

4.2 Magnet Shape

4.2.0.1 Choice: H-shape.

An H-shape magnetic circuit is adopted for the preliminary design. This configuration provides a mechanically robust structure and is well suited for maintaining geometric stability over long-term operation, which is beneficial for field quality. In addition, for the present aperture and yoke sizing constraints, the H-shape offers an efficient and practical layout for coil integration and fabrication.

4.3 Aperture height h

Given that vertical good-field region is $\pm 23\,\mathrm{mm}$ (total $46\,\mathrm{mm}$) we add a vertical clearance aperture of $1mm$ for each pole, and considering a vacuum pipe width of $1\,\mathrm{mm}$, we find:

$$h = (46\,\mathrm{mm} + 2 \cdot 1\,\mathrm{mm} + 2 \cdot 1\,\mathrm{mm}) = 50\,\mathrm{mm}.$$

4.4 Yoke and effective length

The longitudinal dimension is constrained by the physical size available for each magnet, $l_{\text{yoke}} \leq 0.9\,\mathrm{m}$.

By initially tweaking the parameters, we can choose $l_{\text{yoke}} = 0.650\,\mathrm{m}$. For a particle traveling through the magnet, the effective length of the magnet is given by

$$l_{\text{eff}} \approx l_{\text{yoke}} + h = 0.70\,\mathrm{m}.$$

4.5 Flux density B

The bending radius r_m is found by taking to account that the particle bending angle through the magnet is $\theta = 36° = 0.628$ rad, and its cord length is l_{eff}:

$$\begin{aligned} r_m &= \frac{l_{\text{eff}}}{2\sin\theta/2} \\ &= \frac{0.75\,\text{m}}{0.618} = 1.21\,\text{m}. \end{aligned}$$

The calculation for the magnetic rigidity for a particle of charge q, with a rest energy E_0, and a kinetic energy E_k, is given by:

$$B\rho = \frac{p}{q} = \frac{1}{qc}\sqrt{E_k^2 + 2E_kE_0}.$$

Given the following identity,

$$B = \frac{B\rho}{r_m},$$

it is possible to find the magnetic flux density for both protons and Carbon ions to be transferred to the synchrotron.

4.5.0.1 Protons ($q = e$)

$$\begin{aligned} B\rho_p &= 0.383\,\text{T m} \\ B_p &= \frac{0.383\,\text{T m}}{1.21\,\text{m}} \approx 0.31\,\text{T} \end{aligned}$$

4.5.0.2 C^{6+} ($A = 12,\ q = 6e$)

$$\begin{aligned} B\rho_{C^{6+}} &= 2 \cdot 0.383\,\text{T m} = 0.766\,\text{T m} \\ B_{C^{6+}} &= \frac{0.766\,\text{T m}}{1.21\,\text{m}} \approx 0.63\,\text{T} \end{aligned}$$

4.6 Pole Width and yoke width

The total pole width of the magnet is found by taking into account the horizontal Good Field Region (GFR), the sagitta s arising from the curvature of the particle inside the straight magnet, and the optimized pole overhang x_p to ensure minimal pole errors disturbing the fundamental field in the GFR:

$$w = 2x_p + \text{GFR} \times 1.1 + s,$$

with

$$\begin{aligned} x_p &= \frac{h}{2} f\left(\frac{\Delta B}{B_0}\right)_{\text{opt}}, \\ &= \frac{h}{2}\left\{-0.14\ln\left(\frac{\Delta B}{B_0}\right) - 0.25\right\} \\ &= 17.93\,\text{mm} \end{aligned}$$

and the saggitta,

$$s = r_M\left(1 - \cos\left(\frac{\theta}{2}\right)\right) = 59.39\,\text{mm}.$$

Numerically,

$$w = 2 * 17.93\text{mm} + 66\text{mm} + 59.39\text{mm} = 155.25\text{mm}.$$

In order to find the width of the leg, it is sufficient to integrate the magnetic flux through a surface perpendicular to the pole face that encompasses the flux through the yoke, remembering that the flux is shared between both legs of the yoke, as follows

$$\Phi_{\text{gap}} = \Phi_{yoke},$$

where,

$$\Phi_{\text{gap}} = B_{\text{gap}} w_{\text{eff}} = B_{\text{gap}}(w + 2h)$$
$$\Phi_{yoke} = 2B_{\text{yoke}} w_{\text{leg}},$$

which implies ,

$$w_{\text{leg}} = \frac{B_{\text{gap}}(w + 2h)}{2B_{\text{yoke}}}$$

The yoke material is steel M1200, whose saturation occurs at 1.5T, it is a good practice to take a safety 10% margin in design flux with $B_{leg} \approx 1.4\,\text{T}$, hence

$$w_{leg} = \frac{0.63}{2.8}(w + 2h) = 75.26 \text{ mm},$$

4.7 Excitation current per pole

By integrating the Ampère-Maxwell equation,

$$NI = \int_{s_1} \frac{B}{\mu_{\text{air}}} ds + \int_{s_2} \frac{B}{\mu_{\text{iron}}} ds = \frac{Bh}{\mu_0} + \frac{B\lambda_{\text{iron}}}{\mu_{\text{iron}}}$$

Assuming that for unsaturated iron, $\mu_{\text{iron}} \gg \mu_{\text{air}}$, the result simplifies to

$$NI \approx \frac{Bh}{2\mu_0}$$

4.7.0.1 Protons ($B = 0.31\,\text{T}$)

$$NI_p = \frac{0.31 \cdot 50}{2 \cdot 4\pi \times 10^{-7}} = 6167.25\,\text{A turn}$$

4.7.0.2 C^{6+} ($B = 0.56\,\text{T}$)

$$NI_{C^{6+}} = \frac{0.63 \cdot 50}{2 \cdot 4\pi \times 10^{-7}} = 12\,511.93\,\text{A turn}$$

Given that the design specifications must be capable of producing the maximum magnetic flux density required, as in the case of carbon ions, the following calculations therefore use that value as the reference for all subsequent analyses.

4.8 Conductor physical dimensions

From the different available conductors, we present the different physical properties in 3.

The constraint on the current from the converter is $I_{\text{max}} = 600\text{A}$. This imposes a lower bound on the number of turns per pole $N_{\text{min}} = 24$, where each turn carries a current of $I = 521.33$ A, this scenario accounts for a total length of conductor per pole

Table 3: Copper conductor types available on stock (for water-cooled coils).

Type	Copper grade	Outer dims. [mm]	Cooling duct [mm]	Avail. length [m]
1	OF-Cu	25×25	$1{,}20 \times 10^1$	$\approx 3{,}00 \times 10^2$
2	OF-Cu	11×11	$5{,}50 \times 10^0$	$\approx 6{,}00 \times 10^2$
3	OF-Cu	11×11	$5{,}00 \times 10^0$	$\approx 8{,}00 \times 10^2$
4	OF-Cu	8×8	$5{,}00 \times 10^0$	$\approx 5{,}00 \times 10^2$

$$l_{T,\text{pole}} = l_{\text{turn}} \times N$$

where

$$l_{\text{turn}} = 2 \cdot l_{\text{yoke}} + 2 \cdot \pi \cdot P + 2(p_{\text{w}} - 2\rho)$$

with the following quantities:

$$\rho = 3w_c \ ,$$
$$P = \rho + 2w_c \ ,$$

where p_W is the pole width, w_c is the outer diameter of the conductor, ρ the radius of curvature of the nearest conductor to the pole, and P the average radius path for the conductor in the curvatures around the pole taken at a stacking of 4 conductors in the horizontal direction.

The values are tabulated in 4 and 5, it is worth noting that it has been added a total 20% of the values to take into account for the nonlinear behaviour of the path differences between the inner bends and the outer bends, as well as the outer supplies, etc.

Table 4: Length of one turn

Conductor type	l_{turn} [m]
1	2.69
2	2.26
3	2.26
4	2.16

Table 5: Total conductor length per pole and per magnet

Conductor type	$l_{T,\text{pole}}$ [m]	$l_{\text{T,magnet}}$ [m]
1	64.65	129.31
2	54.29	108.59
3	54.29	108.59
4	52.07	104.15

As an initial calculation, the coil envelopes are determined within the available physical space in the horizontal, vertical, and longitudinal directions. A 2 mm insulation layer has been assumed on each side of the conductors for this estimation an a 1 mm insulation between the cables. The values are tabulated in 6.

The overall vertical size of the coil is therefore

$$L_{\text{v}} = 4\, l_{\text{Cu}} + t_{\text{ins,v}}.$$

while the total transversal thickness is given by

Table 6: Coil envelope sizes (as computed) for each conductor option.

Conductor type	Vertical [mm]	Transversal [mm]	Longitudinal [mm]
1	107	159 ,6	1034
2	51	75,6	866
3	51	75,6	866
4	39	57,6	830

$$L_{\mathrm{t}} = 6\, l_{\mathrm{Cu}} + t_{\mathrm{ins,t}}.$$

The longitudinal coil size is computed as:

$$L_{\mathrm{long}} = L_{\mathrm{yoke}} + 3 \cdot w_c \cdot 2 + 2\, L_{\mathrm{t}}.$$

Given that the available longitudinal space for each magnet is 0.9 m, the first conductor would require further parameter tuning. Since it must produce a higher magnetic flux density, it also requires a higher number of ampere-turns. Therefore, we continue the selection considering options 2, 3, and 4.

4.9 Current Density and Power Dissipated

Table 7: Copper cross-section area a_c (excluding the cooling duct).

Conductor type	a_c [mm^2]
1	5.12×10^2
2	9.72×10^1
3	1.01×10^2
4	4.44×10^1

Given the cross section areas for each type of conductor 7, it is possible to determine the resistance of the coil, tabulated in 8, by following the following equation, and using the value for resistivity of Oxygen-Free Copper at 20° is given by $\rho_{\mathrm{Cu}} = 1.71 \times 10^{-8}\ \Omega$ m.

$$R_{\mathrm{coil}}(T_0) = \frac{L_{\mathrm{tot}}\, \rho_{\mathrm{Cu}}}{a_c} \qquad \text{(per pole, at 20°)}$$

By taking into account thermal variations in the resistance per coil, assuming that the average temperature of operation of the coil is taken to be from 20° as the initial working temperature, $T_{avg} = \frac{T_{\mathrm{in}}+T_{\mathrm{out}}}{2}$

$$\begin{aligned} R(T)_{\mathrm{coil}} &= R(T_0)\,(1 + \alpha(T_{avg} - T_0)) \\ &= R(T_0)\left(1 + \alpha\left(\frac{\Delta T}{2}\right)\right) \\ &= \frac{L_{\mathrm{tot}}\rho_{\mathrm{Cu}}}{a_c}\left(1 + \alpha\left(\frac{\Delta T}{2}\right)\right) \end{aligned}$$

The constraint on the maximum increase in temperature is $\Delta T = 15°$, in order to have a 10% safety margin, the maximum allowed increase in temperature will be $\Delta T = 13.5°$. Defining $\alpha = 3.93 \times 10^{-3}/K$ [2] the temperature coefficient of the resistivity of the conductor material. In operation, the inlet water temperature $T_{\mathrm{in}} = 20°$, and

Using Ohm's law, the tension required to provide for the given current to the number of turns is found, it is also possible to add a factor that accounts for the inductive losses in the coil during the ramp-up. Given

Table 8: Coil resistance $R_{\text{coil}}(T_0)$ (per coil).

Conductor type	$R_{\textbf{coil}(T_0)}$ **[mΩ]**	$R_{\textbf{coil}}(T)$ **[mΩ]**
1	2.11	2.16
2	9.75	10.0
3	9.34	9.60
4	20.7	21.26

that the ramp-up time is of the order of seconds, we can assume that the magnet operation is purely DC, such that

$$V_{\text{Dip}} = \left(I\, R_{\text{coil}} + L_{\text{mag}} \frac{\Delta I}{\Delta T} \right) \simeq I R_{\text{coil}}$$

The dipoles are connected in series, such that the total tension is given by the following equation

$$V_{\text{T}} = 3 \cdot V_{\text{mag}}$$

the respective tensions per conductor choice are tabulated in 9

Table 9: Total converter voltage V_{tot} for **3 magnets in series** (computed values).

Conductor type	$V_{\textbf{tot}}$ **[V]**
1	6.78
2	31.29
3	30.02
4	66.51

As the constraint on the voltage is the maximum converter voltage for the three magnets in series is of 100 V it's possible to see that the required tension is below the safety margin of 10% (<90V) for all the conductor types, with a subtle advantage for conductor type 3, which in turn will provide the least amount of ohmic losses.

The total power dissipated by the coil follows Kirchoff's law, and is tabulated in 10

$$P = I\, V_{\text{tot}},$$

Table 10: Electrical power dissipated per **pole** and for **the 3 magnets**.

Conductor type	P **[kW]**	P_{tot} [kW]
1	0.59	3.54
2	2.72	16.32
3	2.61	15.66
4	5.78	34.68

Another key aspect in the power flowing through the coils is the current density, tabulated in 11 by following the equation

$$J = \frac{I}{a_c} \quad [\text{A/mm}^2]$$

From these computations, it is clear that most of the type of conductors require a moderate/high current density ($J > 2$ A/mm^2), this requires water cooling as air cooling is not enough.

Table 11: Current density J (A/mm^2).

Conductor type	J **[A/mm^2]**
1	1.02
2	5.36
3	5.14
4	11.7

4.10 Cooling

The removal of heat by a cooling fluid can be expressed through the energy balance

$$\frac{dW}{dt} = Q\,\delta\,c_p\,\Delta T, \tag{1}$$

where W denotes the thermal energy, Q is the volumetric flow rate of the coolant, δ is its mass density, and c_p is its specific heat capacity.

For a hollow conductor subjected to continuous ohmic heating P_Ω, and assuming steady-state operation with constant heat transfer from the inner conductor wall to the coolant, Eq.(1) becomes

$$Q = \frac{P_\Omega}{\delta\,c_p\,\Delta T}. \tag{2}$$

Given that the current density calculations indicate that the conductors require water cooling, the heat-removal analysis is performed using the thermophysical properties of water at 310 K. By substituting the corresponding values of density $\delta = 993\,\mathrm{kg\,m^{-3}}$, specific heat capacity $c_p = 4.18 \times 10^3\,\mathrm{J\,kg^{-1}\,K^{-1}}$, and kinematic viscosity $\nu = 6.982 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$ into the previous relations, the cooling equations can be rewritten as

$$Q[\mathrm{L\,min^{-1}}] = 14.5\,\frac{P_\Omega[\mathrm{kW}]}{\Delta T[\mathrm{K}]}, \tag{3}$$

The cooling system is evaluated using an inlet water temperature $T_{\mathrm{in}} = 20\,^\circ\mathrm{C}$ and a maximum allowable temperature rise in the magnet of $\Delta T_{\mathrm{max}} = 15\,\mathrm{K}$.

The values are tabulated in table 12 In addition, the available cooling-water pressure is $P_{\mathrm{in}} = 10\,\mathrm{bar}$ and

Table 12: Water flow rate

Conductor type	$Q[\mathrm{L\,min^{-1}}]$
1	0.66
2	2.95
3	2.83
4	6.21

the required return pressure is $P_{\mathrm{out}} = 2\,\mathrm{bar}$, which imposes a practical pressure-drop limit of $\Delta P \leq 8\,\mathrm{bar}$. A lower bound of $\Delta P \geq 1\,\mathrm{bar}$ is also adopted, since excessively small pressure drops may require a more complex and expensive coil geometry. To ensure reliable operation, the coolant velocity must remain below $3\,\mathrm{m\,s^{-1}}$ to avoid erosion and vibration, while the flow should be turbulent ($Re > 4000$) for effective heat transfer. Finally, the Reynolds number must remain within the validity range of the Blasius correlation, $4000 \leq Re \leq 10^5$. The following tables summarize the hydraulic verification of the four conductor candidates according to these criteria.

Using practical engineering units, the average coolant velocity is written as

$$v_{\mathrm{avg}}[\mathrm{ms^{-1}}] = 16.67\frac{Q}{a_h[\mathrm{mm^2}]}, \tag{4}$$

which, for a circular cooling channel, may also be written as

$$v_{\text{avg}}[\text{ms}^{-1}] = 66.67 \frac{Q}{\pi (d_h[\text{mm}])^2}. \tag{5}$$

For a circular pipe, the hydraulic diameter is equal to the inner diameter,

$$d_h = d. \tag{6}$$

The values are tabulated in 13. The pressure drop along the cooling channel is estimated using

Table 13: Average coolant velocity

Conductor type	$v_{\text{avg}}[\text{ms}^{-1}]$
1	9×10^{-2}
2	2.07
3	2.40
4	5.27

$$\Delta P[\text{bar}] = 53.32 \, \frac{Q^{1.75}}{d_h^{4.25}} \, l_h[\text{m}], \tag{7}$$

where $l_h[m]$ is the hydraulic length of the cooling path per pole, since the cooling system will be set to parallel for the two poles. The values are tabulated in 14.

Table 14: Pressure drop

Conductor type	ΔP **[bar]**
1	1.27×10^{-2}
2	5.87
3	8.58
4	32.4

The Reynolds number is evaluated from

$$Re = \frac{v_{\text{avg}} \, d_h}{\nu} \times 10^{-3}, \tag{8}$$

where the factor 10^{-3} accounts for d_h being expressed in mm. The values are tabulated in 15.

Table 15: Reynolds number

Conductor type	**Re**
1	1.79×10^3
2	1.73×10^4
3	1.82×10^4
4	4×10^4

From the hydraulic assessment, only candidates 2 and 3 satisfy all the cooling constraints simultaneously. Candidate 1 is excluded because the flow remains non turbulent ($Re < 4000$) and the pressure drop is far below the target range. Candidate 4 is excluded because both the coolant velocity and the pressure drop exceed the admissible limits.

Among the feasible options, conductor type 3 is selected as the preferred configuration. In addition to satisfying the hydraulic constraints on average velocity and Reynolds number, type 3 exhibits a lower ohmic power loss P_Ω than type 2. This reduces the thermal power that must be removed by the cooling circuit and therefore lowers the required coolant flow rate for a fixed temperature rise ΔT.

However, the calculated pressure drop $\Delta P = 8.58\,\mathrm{bar}$ slightly exceeds the prescribed design limit. To compensate for this, the cooling system can be equipped with two pumps operating in parallel. This configuration allows the required mass flow rate to be maintained while effectively reducing the hydraulic load per pump and ensuring reliable operation within acceptable pressure margins.

As a result, type 3 still provides the most favorable compromise between electrical efficiency, thermal performance, hydraulic feasibility, and operational reliability, with $v_{\mathrm{avg}} = 2.40\,\mathrm{m\,s^{-1}}$ and $Re = 18291.87$ remaining within the imposed limits.

5 Numerical simulation and optimization of the magnet

Finite Element Method Magnet (FEMM) was used in order to model the magnetic field inside the magnet. This program helps solving non-linear magneto-static problem, which is the case for this study. It includes both an automatic mesher and a solver, and the non-linear behavior of the magnetic field inside the yoke with regard to the excitation current was taken into account with saturation curves.

5.1 Simulation of the initial design

A first version of the magnet derived from the calculation above is simulated here. For the H-shape, a quarter of the cross section is modeled. It contains both a quarter of the yoke and a coil.

5.1.0.1 Methodology

The following methodology for the simulation of a magnet in FEMM was adopted :

- **STEP 1 :** Recreate the magnet with the right dimensions and parameters
- **STEP 2 :** Use the automatic mesher to mesh our geometry
- **STEP 3 :** Use the solver and the analysis tool to get the field lines.

5.1.0.2 Initial design performances

The unoptimized magnet gives the following results

Figure 1: Fields lines of the unoptimized magnet

By extracting the magnetic flux values inside the GFR, we can draw the relative error on magnetic field along the GFR :

It is clear that this unoptimized version does not fulfill the $\frac{\Delta B}{B_0} < 10^{-3}$ requirement on the GFR.

5.2 Optimization

5.2.0.1 Methodology

The optimization part will focus on shimming around the poles in order to concentrate the field lines between the poles, to enhance the GFR $\frac{\Delta B}{B_0}$ values and to minimize stray field in the coils. Some part of the yoke that had the lowest magnetic field values were also cut out. The final design was obtain by an iterative process, adding and removing steel and analyzing the effect on the value of the $\frac{\Delta B}{B_0}$.

Figure 2: $\frac{\Delta B}{B_0}$ in the center of the good field region for the unoptimized magnet

5.2.0.2 Optimized magnet

With this optimized design of the magnet, the magnetic field lines were enhanced, and the value of the relative magnetic field error was kept above 10^{-3}.

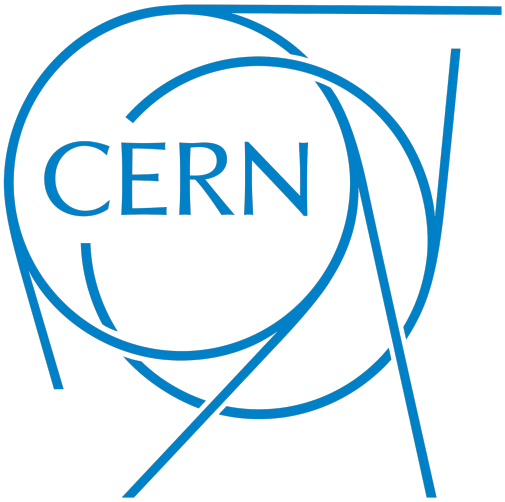

Figure 3: Cross section of the optimized magnet

Figure 4: $\frac{\Delta B}{B_0}$ for the optimized magnet

6 Cost estimate

Costs were divided between fixed costs (design, tools,...), variable costs that scales with the number of magnets, and electric consumption costs, from both magnets and pumps.
The amount of MWh was estimated with the previous calculations, with different values for the potential (V) and the intensity (I) based on the ion we consider (p^+ or C^{6+}).

Table 16: Estimation of energy consumption for magnets

Ion		Value
p+	U (V)	9.08
	I (A)	160.97
	E (MWh)	85.9
C6+	U (V)	31.5
	I (A)	558
	E (MWh)	442.94

Here the magnets are in series, so it only requires one source for the voltage. We considered 20 years of use with a factor of charge of 50%, 70% of the time using protons, 30% using ions.

An estimation of the power consumption for the pumps to flow water through the coil was conducted. With the flow ($Q = 2.8\ L/min$) and the pressure drop ($\Delta P = 8\ bar$), we estimate the power of a pump to be around $100\ W$. With 6 pumps, we calculate a total energy amount of roughly $50\ MWh$.

Based on the area of the magnet cross section, we can access the total mass of copper and steel.

Table 17: Costs estimation

	Cost Category	Cost Factor	Unity cost	Quantity	Total
Fixed costs	Design	Design work	15,000 €	1	15,000€
	Process	Stamping tool	15,000 €	1	15,000€
		Stacking tool	12,000 €	1	12,000€
		Winding tool	10,000 €	1	10,000€
		Impregnation mould	18,000 €	1	18,000€
Variable costs	Material	Steel	1.80 €/kg	3250.76 kg	5,851.36€
		Copper conductor	18.00 €/kg	193.12 kg	3,476.22€
	Manufacturing	Yoke	12.00 €/kg	3250.76 kg	39,009.05€
		Coil	36.00 €/kg	193.12 kg	6,952.43€
	Auxiliary parts	Auxiliary parts	5,000 €	4	20,000€
	Electricity	Coils	95 €/MWh	528,88 MWh	50,243.88€
		Pumps	95 €/MWh	50.40 MWh	4,788.00€
				Total	200,320.94€

Bibliography

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Appendix

A Title of appendix

A.1 Subsection title in appendix